

Quest Pro enhanced face tracking

V2.1.0 release candidate | USB or wireless ADB | 1 October 2026

A rooted Meta Quest Pro and the latest VRCFaceTracking from Steam are required. If you used a version before V2.0, record and train a new tongue model. Working V2.0 through V2.0.2 models can be exported and imported into V2.1.0. Developer v8 remains the default. Mustachio is optional and highly experimental.

Based on [Qpro-Enhanced-FT by n0tmast3r](#). Fwooffy maintains the [AMD/NVIDIA wireless edition](#). Fwooffy wrote the original [Quest Pro root guide](#); glorpette made and hosts the linked GitHub version.

Start with the headset's eye tracking, then set up the PC software. The **Hub** is the QproFaceTracking.exe program. **ADB** is the connection it uses to talk to your Quest. The **Qpro module** sends tracking results to VRCFaceTracking.

1. Set up eye tracking and independent gaze first

1. On the rooted Quest Pro, open **Settings > Movement Settings**. Turn on **Eye Tracking** and **Natural Facial Expressions**, then run eye calibration if offered. If these are already enabled, leave them on. Keep Developer Mode enabled.
2. If **Magisk OverlayFS** is not installed yet, open **Singularity > Apps/Modules > Magisk repo** and install it. Follow any reboot prompts. If it is already enabled in Magisk, you can skip reinstalling it.
3. In the same Singularity menu, install **Quest Pro Independent Eye Gaze** if it is not installed yet. Reboot if prompted, then check in Magisk that both modules are enabled. This is the headset-side independent gaze route.
4. Once your selected headset streaming app and VRCFaceTracking are installed in section 3, check that left and right gaze and convergence move correctly. The Magisk route has been reported working on **Horizon OS v2.7**, but exact firmware builds can behave differently.

Use one independent gaze method at a time. While the Magisk module is active, leave **Independent Eye Gaze** unchecked in the Hub. If you want to try the Hub's temporary gaze method instead, enable it and start tracking; the headset will freeze while it hooks into the tracking engine. **Stop tracking** only reverses the Hub's changes; it does not disable a Magisk module.

2. Download and open the Hub

1. Download **QproFaceTracking V2.1.0.zip** when it is published on [this project's Releases page](#), or use the supplied release candidate ZIP. Choose the named QproFaceTracking ZIP, **not** GitHub's "Source code" ZIP.
2. In Windows File Explorer, right-click the ZIP and choose **Extract All**. Open the extracted folder. Keep QproFaceTracking.exe, Helpers, and QproRuntime together.
3. Double-click QproFaceTracking.exe. It opens on **First-time setup** the first time. You can return to that page from the menu on the left.

You need an internet connection and several gigabytes of free space for setup. The ZIP already includes ADB. The Hub installs the Python parts it needs, so you do not have to install ADB or Python separately for this program.

3. Get the PC apps ready

If Magisk asks whether **Shell / ADB Shell** may have root access, allow it. Keep the Quest awake while setting up the Hub.

Choose either [Virtual Desktop and its PC Streamer](#) or Steam Link for your Quest-to-PC VR connection. Install [SteamVR](#) and the [latest VRCFaceTracking from Steam](#) on the PC. **The current Steam VRCFaceTracking is required for either source.** Let Steam finish updating it before installing Qpro's module. After installing the matching Qpro module in section 5, try ordinary face and blink tracking before turning on Qpro's extra features.

Steam Link settings

You can choose **Steam Link** under **Streaming app** instead of Virtual Desktop. SteamVR and the latest VRCFaceTracking are still required. In Steam Link on the headset, open **Advanced Settings**, turn on **OSC**, **Share eye tracking data**, and **Share face tracking data**, then set **OSC Output Port** to **9015**. These are Steam Link's tracking settings; the Hub's **Connection type** still chooses USB or wireless ADB for Qpro's headset camera stream.

Close VRCFaceTracking and wait for its module process to exit before switching sources. In **First-time setup**, choose **Steam Link** and press **Install Steam Link module**; reopen VRCFaceTracking afterward. Installing this Qpro module removes the Qpro Virtual Desktop module. A separate LinkFT or Steam Link VRCFaceTracking module still needs to be removed through VRCFaceTracking because it competes for the same OSC port and eye/face slots. Steam Link face, blink, and mouth tracking worked after a headset restart, Qpro's camera tongue override moved in VRChat, and native Steam TongueOut worked with VRCFaceTracking running on Steam Link. To return to Virtual Desktop, choose it in the Hub, press **Install Virtual Desktop module**, then reopen VRCFaceTracking.

If you want to stream the **Virtual Desktop picture over a USB cable**, Fwooffy's working setup used the Virtual Desktop **Beta** channel. In the Meta Horizon phone app or the headset's app library, long-press Virtual Desktop, open **Settings > Release Channels**, choose **Beta**, and update it. This changes VR video streaming. The Hub's USB or wireless choice below controls **tracking data**, which is separate.

4. Connect your Quest to the Hub

Choose **one** connection type on **First-time setup**. If this is your first time using the Hub, USB is the easiest way to check that it works. You can switch to wireless later.

Option A: USB cable

1. Plug the Quest Pro into the PC and put the headset on.
2. If a **USB debugging** message appears in the headset, choose **Allow**.
3. In the Hub, set **Connection type** to **USB cable**. If the Hub says it can see the headset but has no root access, open Magisk on the Quest and allow **Shell / ADB Shell**. Then use **Refresh connection status** on the Live tracking page.

Option B: Wireless ADB

1. Connect the Quest and PC to the same private Wi-Fi network. On **First-time setup**, choose **Wireless ADB (Wi-Fi)**.
2. Enter the Quest's Wi-Fi address in **Quest IP:port**. It usually looks like 192.168.1.25:5555. The numbers before :5555 are *your Quest's* address, so do not copy that example exactly. Press **Connect to Quest**.
3. A **Quest connected** pop-up means wireless ADB and Magisk root are ready. Activity will also say **Wireless Quest ready**. **Do not press Pair and connect**. You can unplug the USB cable, if one was attached. If a **Quest connection failed** pop-up appears, check the Quest's current IP and port, keep it awake, and read **Activity** for the exact error.
4. If the wireless connection is off but USB works, plug in USB once and press **Enable from USB**. After the Hub reports success, unplug USB and connect to the saved Quest address.

Pair and connect is only for a headset that shows a **six-digit wireless debugging pairing code**. Keep the pairing message open in the headset. Put the short-lived address shown next to the code into **Pairing IP:port**, put the code into **Six-digit code**, and keep the normal Quest address in **Quest IP:port**. The two port numbers are different. For example, 192.168.1.25:5555 is a normal connection address, **not** a pairing address. Use your Quest's actual address.

If pairing says `protocol fault`, open a new pairing message on the headset and use its new code and pairing port. If **Connect to Quest** already said ready, skip pairing. Your Quest's Wi-Fi address may change after a reboot; update it in the Hub if it does. Use wireless ADB only on a trusted network.

5. Press the setup buttons

On **First-time setup**, work down the three numbered cards:

1. On the **PC runtime** card, press **Install runtime** and wait for Activity to say it has finished. This prepares Qpro's private Python and downloads the PC parts needed for tracking. The Hub stays at your current place on the page while it runs. You do not need to install, repair, or remove another Python installation.
2. **Close VRCFaceTracking** and wait for its module process to exit. Choose your **Streaming app**, press the matching **Install Virtual Desktop module** or **Install Steam Link module** button, wait for completion, then reopen VRCFaceTracking. Only the selected app's install button is available. Installing one Qpro module uninstalls the other Qpro source module.
3. If you chose the Hub's independent gaze method **instead of the Magisk module**, press **Prepare gaze** while the rooted Quest is connected. The Hub reads the stock eye model and tracking-engine details from that headset, then prepares a temporary model on the PC. You do not need to supply an engine file. Skip this button if the Magisk gaze module is enabled, or if you only want tongue or pupil tracking. Prepare gaze again after a headset firmware update.

The Hub checks the prepared model and headset build before restarting tracking. If the build is unsupported or changed, Activity explains why and the Hub leaves headset tracking alone. Use the Singularity module instructions in section 1, or keep ordinary eye tracking from your selected source. Note your Quest's **exact firmware build** when asking for help. On a supported build, applying or restoring the Hub's temporary eye model restarts Meta trackingservice. The headset may briefly look frozen and lose positional tracking; this is an expected restart, not a headset crash. Wait for Activity to confirm that tracking has returned.

To remove Qpro's VRCFaceTracking add-on later, close VRCFaceTracking, wait for its module process to exit, and press **Uninstall Qpro module** on the **VRCFT module** card. Confirm the prompt, wait for **Setup step complete**, then restart VRCFaceTracking. The Hub removes Qpro's installed source module and restores any Virtual Desktop modules this extracted copy saved during installation. It does not delete your tongue recordings or models. If ordinary Virtual Desktop face tracking is missing afterward, install its official VRCFaceTracking module again.

AMD GPU: After **Install runtime** succeeds, press **Install ROCm 10.0** if the Hub detects one of these discrete Radeon models on Windows 11. If an older verified ROCm 7.2.1 environment is already present, the button says **Upgrade to ROCm 10.0** instead:

Series	Models mapped for ROCm 10.0
RX 6000	6950 XT, 6900 XT, 6800 XT, 6800; 6750 XT, 6700 XT, 6700; 6650 XT, 6600 XT, 6600
RX 7000	7900 XTX, 7900 XT, 7900 GRE; 7800 XT, 7700 XT, 7700; 7600 XT, 7600
RX 9000	9070 XT, 9070, 9070 GRE; 9060 XT, 9060
Radeon PRO	AI PRO R9700, PRO W7900, PRO W7900 Dual Slot

Qpro installs [AMD TheRock ROCm 10.0 packages](#) for the detected card's [GPU target](#). These are AMD's stable packages, while their use in Qpro remains **experimental**. AMD's package index includes Windows device packages for the [RX 6750/6700 family \(gfx1031\)](#) and [RX 6650/6600 family \(gfx1032\)](#). Package availability does not guarantee driver compatibility on every card. Confirm the Hub prompt and allow time for a large download. [AMD's ROCm 10.0 matrix](#) validates Windows 11 25H2 with Adrenalin 26.8.1; other Windows 11 builds or drivers may fail the GPU checks. Qpro enables the new environment only after GPU training and model inference tests pass. Integrated graphics and unlisted Radeon models are not eligible. If setup fails, read the full **Activity** error.

The new packages use `.venv-rocm-experimental`. This folder name preserves compatibility with earlier test builds; it does not mean AMD's packages are a preview release. If this extracted Qpro copy already has a **verified** `.venv-rocm` installation on a card in [AMD's Windows ROCm 7.2.1 list](#), Qpro keeps it as a fallback when ROCm 10.0 is not ready. The Hub shows whether 10.0 is verified or the older fallback is available. An RX 7900 XTX was live-tested with the older ROCm 7.2.1 path; the ROCm 10.0 Qpro path still needs live tests on the listed cards.

NVIDIA GPU or CPU: skip the AMD button. NVIDIA CUDA hardware has been live-tested and is functional. When tracking starts, **Activity** tells you whether the tongue model is using AMD, NVIDIA, or CPU.

6. Record and train your tongue model

Train if you have no personal model or used a version before V2.0. You can import a working V2.0 through V2.0.2 model into V2.1.0. The bundled models were trained on other wearers.

1. For Quick refinement or Focused training, choose the model you want to extend under **Live tracking > Tongue model** before training. Developer v8 is the default. Full dataset makes a new personal model instead.
2. Open **Personalize** and choose a capture mode below.
3. Follow each pose card, hold the pose steady, and press **SPACE** once per still. Save at least the on-screen minimum for every card before pressing **ENTER** to continue. Include the side and **diagonal** positions; missing a pose can leave a tracking gap there.
4. When recording finishes, choose it in the training list and press that mode's training button. Wait for Activity to say training finished. Select the resulting model when you restart tracking in section 7.

- **Quick refinement - 10-20 min:** press **1. Record refinement**, then **2. Train personalized copy**. This is the shortest way to adapt the selected model to your face.

- **Focused diagonals + facial hair - 15-30 min:** press **1. Record focused dataset**, then **2. Train focused copy**. Its 22 cards add diagonal positions and matched hidden/visible poses for facial-hair shadows.

- **Full dataset - 60-120 min:** press **1. Record full dataset**, then **2. Train new personal model**. Its 58 cards cover more mouth poses, directions, and tongue extension strengths. Take breaks as needed.

If you have facial hair or wear a bandage near the mouth, keep it as you normally wear it during tracking. For every visible pose, check that the tongue can actually be seen in the two camera panels. Hold one pose per still; vary your jaw or headset position slightly between stills. Record hidden poses too, so beard shadows and smiles are not mistaken for tongue movement. Hair that completely covers the tongue cannot be removed by training or contrast processing. The capture screen waits for fresh VRCFaceTracking factory packets; it does not require the native face values to move before recording.

Trying Mustachio

Mustachio is highly experimental and optional. It extends developer v8 with a focused capture from one bearded and moustached wearer. Independent clean-shaven testing and broader facial-hair testing are still pending, so it may perform worse for some people. Developer v8 remains available and is the default.

To try it, choose **Mustachio** under **Live tracking > Tongue model**, then restart tracking. Its contrast processing is selected automatically by the model; it does not invent hidden tongue pixels or remove hair. Quick refinement and Focused training can extend this selected model. Those copies retain the **highly experimental** label, including after renaming, export, and import. Choosing Full dataset starts a new personal model.

After training, **Activity** lists the weakest held-out pose cards and diagonal corners. missed counts visible-tongue frames that the selected visibility gate marked hidden; fnr is that count divided by visible frames. These are checks on held-out frames from the same recording, so try the model live as well.

The **Model manager** can export a model as a backup. Import working V2.0 through V2.0.2 exports into V2.1.0; for pre-V2.0 exports, make a new capture and train again. Camera recordings are personal data, so share them only if you want to.

To remove a recording you no longer need, stay on **Personalize**. Under its matching capture card, choose it from **Recorded datasets (including trained)** and press **Delete selected dataset....** Read the confirmation before choosing **Yes**: this permanently removes the recording, its labels and session details, and its prepared training cache from this extracted copy. A tongue model already trained from that recording remains available in **Model manager**. The training dropdown above only lists datasets still waiting to be trained.

7. Start tracking

1. Start your selected **Virtual Desktop** or **Steam Link** app on the Quest, then SteamVR and VRCFaceTracking on the PC. Steam Link users should confirm its OSC and eye/face sharing settings above.

2. In the Hub, open **Live tracking**. Independent Eye Gaze starts off; select it, tongue tracking, pupil dilation, or the combination you want. **If you installed Singularity's Independent Eye Gaze Magisk module, leave the Hub's Independent Eye Gaze option off.** For tongue tracking, choose the **new model you just trained** under **Tongue model**. If tracking is already running, press **Stop tracking** first so the new model loads when tracking restarts.
3. Press **Start tracking**. Open **Activity** if you want to see whether the camera connected and which device is running the tongue model.
4. When you finish, press **Stop tracking**. Wait until Activity says the camera has stopped and the stock eye model has been restored.

Closing the Hub stops tracking cleanly, then stops the configured PC ADB server. Other Android tools using the same ADB server will disconnect and may restart it. Finish setup, capture, training, or eye-model recovery before closing.

Preview tracking cameras lets you hide the camera windows without turning tracking off. Change it before starting the next session. **Pupil response** makes avatar pupil changes stronger or weaker; start near the default and adjust slowly. Pupil values are estimates for avatar animation, **not** measured eye health data. Your VRChat avatar must support pupil animation for changes to appear.

Choose your cheek puff response

Individual cheek puff is on **Live tracking** and starts on with **1/0** selected. Choose a **Cheek puff style**:

- **Calibrated**: a smooth strength from relaxed to full puff, including values between 0 and 1. It uses your personal calibration for the selected streaming app, or the bundled developer cheek baseline until you make one. Separate developer baselines were measured on one Quest Pro through Virtual Desktop and Steam Link. They are a starting point; calibrate for your own face if the response is too weak or too strong.
- **1/0**: a clear one-cheek puff becomes full strength on that side and zero on the other. Puffing both cheeks still moves both.
- **Balanced**: gentler separation that keeps changes in cheek strength without using a calibration profile.

Turn **Individual cheek puff** off to use the original cheek values from Virtual Desktop or Steam Link. The toggle and style take effect while tracking is running. If the VRCFaceTracking preview separates the cheeks but the avatar does not, check the avatar's parameters and blendshapes.

Calibrate cheek puff for your face

1. Keep the headset on and your chosen streaming app connected. Its matching Qpro module must be installed and tracking in VRCFaceTracking. **Qpro camera tracking does not need to be running.**
2. On **Live tracking**, press **Calibrate cheek puff**. Confirm that the window says **Live cheek feed is ready**.
3. Relax both cheeks, press **Capture relaxed cheeks**, and hold still for three seconds.
4. Puff only your own left cheek, keeping the right relaxed. Press **Capture left cheek** and hold for three seconds. Relax briefly afterward.
5. Puff only your own right cheek, keeping the left relaxed. Press **Capture right cheek** and hold for three seconds. Relax briefly afterward.
6. Puff both cheeks together, press **Capture both cheeks**, and hold for three seconds.

The window saves your profile after all four poses pass, then selects **Calibrated** and turns **Individual cheek puff** on. Keep your jaw comfortable and avoid smiling during the holds. The window reads the original cheek signals, so both numbers can rise during a one-cheek puff. Calibration measures that overlap. If a pose is too weak, too noisy, or cannot be distinguished from the other side, follow the message and repeat that step. A paused or mismatched feed must be fixed before you continue. If the window reports an older calibration feed, close VRCFaceTracking, install the matching Qpro module from this build, and reopen VRCFaceTracking before trying again. Closing the window before completion keeps any existing profile.

Virtual Desktop and Steam Link have separate personal profiles. Calibrate each streaming app you use; switching back uses that app's saved profile. This is a short cheek calibration, so you do not need to record or retrain your tongue model.

Individual cheek suck uses the separate left and right cheek-suck signals from Virtual Desktop or Steam Link. It starts on with **Strong individual (1/0)**, which emphasizes the stronger side while leaving a deliberate two-cheek

suck on both sides. Choose **Balanced** for a softer effect, or turn it off to pass through the streaming app's original values. It does not use negative cheek-puff values.

Adjust eyebrow movement starts off, so brows use the selected source's original values. Turn it on and adjust **Eyebrow sensitivity** to amplify or soften the existing left and right inner raise, outer raise, and lower/pinch expressions. This cannot add movements the headset does not detect. The avatar needs matching brow parameters and blendshapes to show that detail. If the expressions move in VRCFaceTracking's preview but the avatar shows only a single brow motion, check the avatar's face tracking setup.

Smirk handling is built into both Qpro modules. A clear one-sided smile emphasizes its leading corner and suppresses the weaker side; an even smile still moves both sides. There is no separate smirk toggle in the Hub.

For tongue motion, adjust **Motion smoothing** to soften extension and retraction. The visibility hold reduces quick dropouts after a visible pose, but it cannot make a completely obscured tongue visible. Use **Focused diagonals + facial hair** to record positions that still need better coverage.

Common problems

If Independent Eye Gaze fails to start or stops unexpectedly, the Hub turns it off automatically and remembers that choice. Tongue and pupil tracking continue if selected. Check **Activity** for the recovery result before trying gaze again; select **Independent Eye Gaze** manually to retry.

What you see	What to try
"Root access unavailable"	Keep the Quest awake. In Magisk, allow Shell / ADB Shell , then refresh the connection.
Quest connected pop-up, followed by a pairing error	You are already connected. Skip Pair and connect and start tracking.
Quest connection failed pop-up	Check the Quest's current Wi-Fi IP and port, wake the headset, and read Activity for the specific error.
protocol fault during pairing	Use a fresh six-digit code and the temporary pairing port shown on the Quest, not the usual :5555 port.
Wireless stopped working after a reboot	Check the Quest's current Wi-Fi address. Wireless ADB may need to be enabled again.
AMD installer says PC runtime is missing	Finish Install runtime first, then return to Install ROCm 10.0 .
ROCm setup reports a missing torchgen or a failed Python import	Use Verify / repair ROCm 10.0 or Repair ROCm 10.0 in First-time setup. Qpro checks and repairs its separate AMD environment before testing the GPU. Keep the full Activity error if repair fails; an import error alone does not mean your card is unsupported.
Install runtime stays on "Starting PC runtime setup"	Keep the Hub open and read Activity . It now shows the PowerShell launch, current setup phase, and a still-running message every 15 seconds when output stops. If PowerShell never prints its first script message, check Windows Security for a blocked powershell.exe, then share the Activity lines, including the PowerShell PID.
Install runtime ends with code 1	Open Activity and read the first error above the exit code. If it mentions a missing DLL, VCRUNTIME, or a Python import failure, install or repair Microsoft's latest x64 Visual C++ Redistributable , then reopen the Hub and run Install runtime again. Code 1 has other causes too, so share the full error if it repeats. Do not delete the whole %LOCALAPPDATA%\QproFaceTracking folder just because the exit code is 1.

What you see	What to try
Steam Link source selected but no face or training labels arrive	In Steam Link Advanced Settings , turn on OSC , Share eye tracking data , and Share face tracking data ; set OSC Output Port to 9015 . Close VRFaceTracking and wait for its module process to exit, then press Install Steam Link module in the Hub. Reopen VRFaceTracking. Remove other Steam Link VRFaceTracking modules that would use the same port.
Launcher says the tracking source does not match	Close VRFaceTracking and wait for its module process to exit. Choose the intended Streaming app on First-time setup , install its Qpro module, reopen VRFaceTracking, then retry.
Calibrate cheek puff is waiting for live values	Keep the headset on and the selected streaming app connected. Start tracking in VRFaceTracking with the matching Qpro module from this build. If the window reports the other streaming app, select the correct Streaming app in the Hub or install its matching module with VRFaceTracking closed. Qpro camera tracking is not required.
AMD Ryzen integrated graphics appears during ROCm setup	Qpro checks for one of the exact discrete Radeon models in the ROCm 10.0 table above. An integrated GPU alone is not enough. Check that Windows and the AMD driver can see the discrete card; otherwise use Install runtime for NVIDIA CUDA or CPU.
Independent gaze or convergence fails but ordinary face tracking works	Check the Quest's exact firmware build. Follow the Singularity Magisk setup in section 1 ; it has been reported working on Horizon OS v2.7.
Tongue disappears in some positions	Check that your newly trained model actually loaded in Activity. Record those positions again and retrain.
Facial hair or a bandage makes tongue training unreliable	Record a personal dataset with the same face appearance you use for tracking. Capture hidden and visible tongue poses, then check the trained model in Live tracking. If native visibility flickers while the cameras still see your tongue, try Camera only under Visibility . This may help but is not guaranteed for every face or headset fit.
Pupils seem too jumpy	Lower Pupil response and hold your gaze steady while tracking warms up.

If Windows will not let you delete an older extracted Qpro folder, close its Hub and camera preview first. If the folder is still in use, open **Task Manager > Details**, end a leftover adb . exe process, then try again. Ending adb . exe temporarily disconnects any other Android tools using that ADB server.

If you need help, copy the relevant lines from **Activity** and tell us your Quest firmware version, whether you used USB or wireless, and which release ZIP you downloaded. You can ask in the [community Discord](#). The server is not owned by Fwooffy.

The Steam Link source follows the OSC mapping documented by [danwillm](#) and the [LinkFT project](#). Qpro does not require either third-party module alongside its own module.

Tested PC example

The owner reported good performance with few issues on Windows 11 Pro, an Intel Core i7-13700KF, AMD Radeon RX 7900 XTX, and 32 GB RAM. This is an experience report, not a benchmark or a guarantee for other PCs.